ANOVA Table (one-way)

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F-ratio
Between treatments	160.14	3 – 1 = 2	80.07	$F_o = \frac{80.07}{8.73} = 9.17$
Error	104.8	12	8.73	
Total	264.94	$n - 1 = 15 - 1 = 14$		

Step 6 : Critical value

$f_{(2, 12),0.05} = 3.8853.$

Step 7 : Decision

As $F_0 = 9.17 > f_{(2, 12),0.05} = 3.8853$, the null hypothesis is rejected. Hence, we conclude that there exists significant difference in the reduction of the average blood pressure in atleast one pair of techniques.

Example 3.5

Three composition instructors recorded the number of spelling errors which their students made on a research paper. At 5% level of significance test whether there is significant difference in the average number of errors in the three classes of students.

Instructor 1	2	3	5	0	8		
Instructor 2	4	6	8	4	9	0	2
Instructor 3	5	2	3	2	3	3	

Solution:

Step 1 : Hypotheses

Null Hypothesis: $H_0 : \mu_1 = \mu_2 = \mu_3$

That is there is no significant difference among the mean number of errors in the three classes of students.

Alternative Hypothesis

$H_1 : \mu_i \neq \mu_j$ for at one pair (i, j); $i,j = 1,2,3$; $i \neq j$.

That is, atleast one pair of groups differ significantly on the mean number of errors.

12th_Statistics_EM_Unit_3.indd 88 1/28/2020 6:33:40 PM